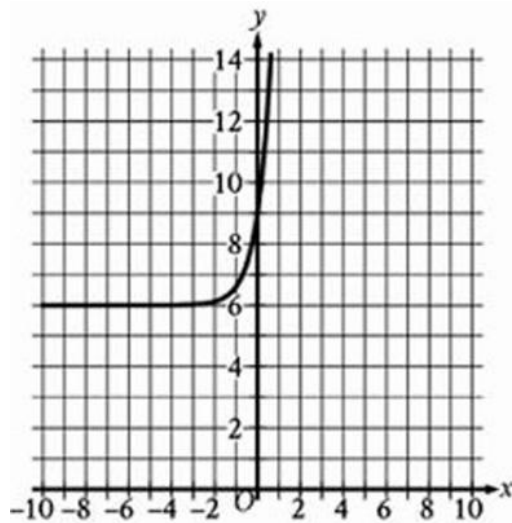


1. $s = 10 + 4t$

The equation gives the speed s , in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car 5 seconds after it began to accelerate?

- A. 10
- B. 14
- C. 15
- D. 30

2. What is the y-intercept of the graph shown?



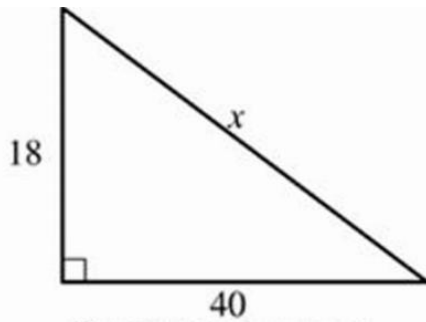
- A. (-9,0)
- B. (-6,0)
- C. (0,6)
- D. (0,9)

3. The width of a rectangle is 5 centimeters. The length of the rectangle is 30 centimeters longer than the width. What is the area of the rectangle, in square centimeters?

- A. 175
- B. 40
- C. 10
- D. 5

4. If $6 + x = 3$, what is the value of $42 + 7x$?

5. In the right triangle shown, what is the value of x ?



Note: Figure not drawn to scale.

- A. $\sqrt{58}$
- B. $\sqrt{116}$
- C. $\sqrt{720}$
- D. $\sqrt{1924}$

6. If $\frac{x}{y} = 87$ and $\frac{cx}{3y} = 87$, what is the value of c ?

7. In right triangle RST, the sum of the measures of angle R and angle S is 90 degrees.

The value of $\sin(R)$ is $\frac{4\sqrt{2}}{9}$. What is the value of $\cos(S)$?

- A. $\frac{4\sqrt{2}}{9}$
- B. $\frac{4\sqrt{2}}{7}$
- C. $\frac{7\sqrt{2}}{8}$
- D. $\frac{9\sqrt{2}}{8}$

8. Antwon's hair will grow at a constant rate from a length of 3 inches to a length of 5 inches. The equation $0.3t + 3 = 5$ describe this situation, where t is the number of months it will take for his hair to reach a length of 5 inches. Which of the following is the best interpretation of 0.3 in this context?
- A. The time, in months, it will take Antwon's hair to grow 1 inch
 - B. The length, in inches, Antwon's hair will grow each month
 - C. The time, in months, it will take Antwon's hair to reach a length of 5 inches
 - D. The length, in inches, Antwon's hair will grow in 5 months
9. Which of the following lists represents a data set with the smallest standard deviation?
- A. 72, 73, 75, 77, 78
 - B. 73, 73, 75, 77, 77
 - C. 73, 74, 75, 76, 77
 - D. 74, 75, 75, 75, 76
10. In the linear function f , $f(0) = 7$ and $f(2) = 7$. Which equation defines f ?
- A. $f(x) = 0$
 - B. $f(x) = 2$
 - C. $f(x) = 7$
 - D. $f(x) = x + 7$
11. The product of a positive number x and the number that is 1 less than x is equal to 272. What is the value of x ?
- A. 0.5
 - B. 17
 - C. 271
 - D. 273
12. What is the positive solution to the given equation?

$$4x^2 - x = 18$$

13. A sample of a certain isotope takes 29 years to decay to half its original mass. The

function $s(t) = 184(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 184-gram sample starts to decay. Which statement is the best interpretation of $s(58) = 46$ in this context?

- A) Approximately 46 grams of the sample remains 58 years after the sample starts to decay.
- B) The mass of the sample has decreased by approximately 46 grams 58 years after the sample starts to decay.
- C) The mass of the sample has decreased by approximately 58 grams 46 years after the sample starts to decay.
- D) Approximately 58 grams of the sample remains 46 years after the sample starts to decay

14. Tanya drove on the highway and on local roads to complete a trip of 290 miles. The drive took 5 hours. She drove an average speed of 65 miles per hour (mph) on the highway and an average speed of 30 mph on local roads. If x is the time, in hours, Tanya drove on the highway, and y is the time, in hours, she drove on local roads, which system of equations represents this situation?

- A. $65x + 30y = 5$
 $x + y = 290$
- B. $65x + 30y = 290$
 $x + y = 5$
- C. $30x + 65y = 5$
 $x + y = 290$
- D. $30x + 65y = 290$
 $x + y = 5$

15. Which point (x, y) is a solution to the given inequality in the xy -plane?

$$y < -2x + 14$$

- A. (0,15)
- B. (-1,17)
- C. (-2,0)
- D. (8,-1)

16. What is the sum of the solutions to the given equation?

$$(x - 35)^2 = 1$$

17. A dance studio charges an introductory fee for the first 3 lessons then charges a fixed fee for each additional lesson. One student took 6 lessons and was charged \$150. Another student took 17 lessons and was charged \$480. Which function f gives the total charge, in dollars, for any student who took x lessons, where $x > 2$?

A. $f(x) = 25x$

B. $f(x) = 25x + 55$

C. $f(x) = 30x$

D. $f(x) = 30x - 30$

18. How many centimeters are equivalent to 35 meters? (1 meter = 100 centimeters)

19. A researcher observes a sample of a nuclide. An exponential model estimates that the mass, in grams, of the sample decreases by 23% every 11.69 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the researcher began observing the sample?

A. $M = 100(0.77)^{\frac{t}{11.69}}$

B. $M = 100(0.77)^{t+11.69}$

C. $M = 100(0.23)^{\frac{t}{11.69}}$

D. $M = 100(0.23)^{t+11.69}$

20. Data sets A and B each consist of 6 values as shown, where h is a constant. If the standard deviation of set A is greater than that of set B, which of the following could be the value of h ?

Data set A: 8, 11, 12, 15, 21, 27

Data set B: 9, 12, 13, 16, 22, h

- I. 23
- II. 28
- III. 29

- A. I only
- B. II only
- C. III only
- D. II or III

21. In triangle RST, the measure of angle R is 50° , the measure of angle S is x° , and the measure of angle T is $(4x - 5)^\circ$. Point L lies on RS, point K lies on ST, and LK is parallel to RT. What is the measure, in degrees, of angle SKL? (Disregard the degree symbol when entering your answer.)

22. The function $f(x)$ is defined by the given equation. If $f(a) = -9a$, where a is a constant, what is the value of a ?

$$f(x) = \sqrt{6x + 7}$$

- A. $-\frac{1}{3}$
- B. $-\frac{7}{27}$
- C. $\frac{7}{27}$
- D. $\frac{1}{3}$